

CPU Scheduling Simulation Report

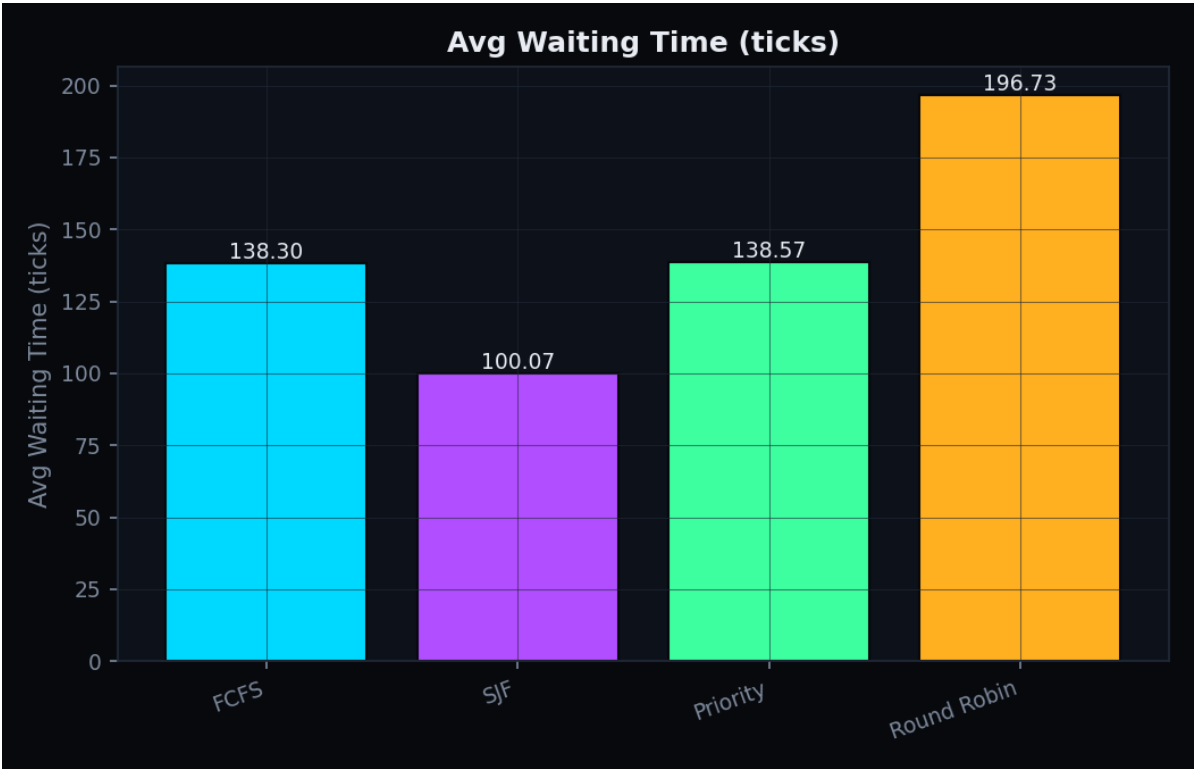
Workload: 30 processes · seed 42 · Round Robin quantum 4

Algorithm Comparison Table

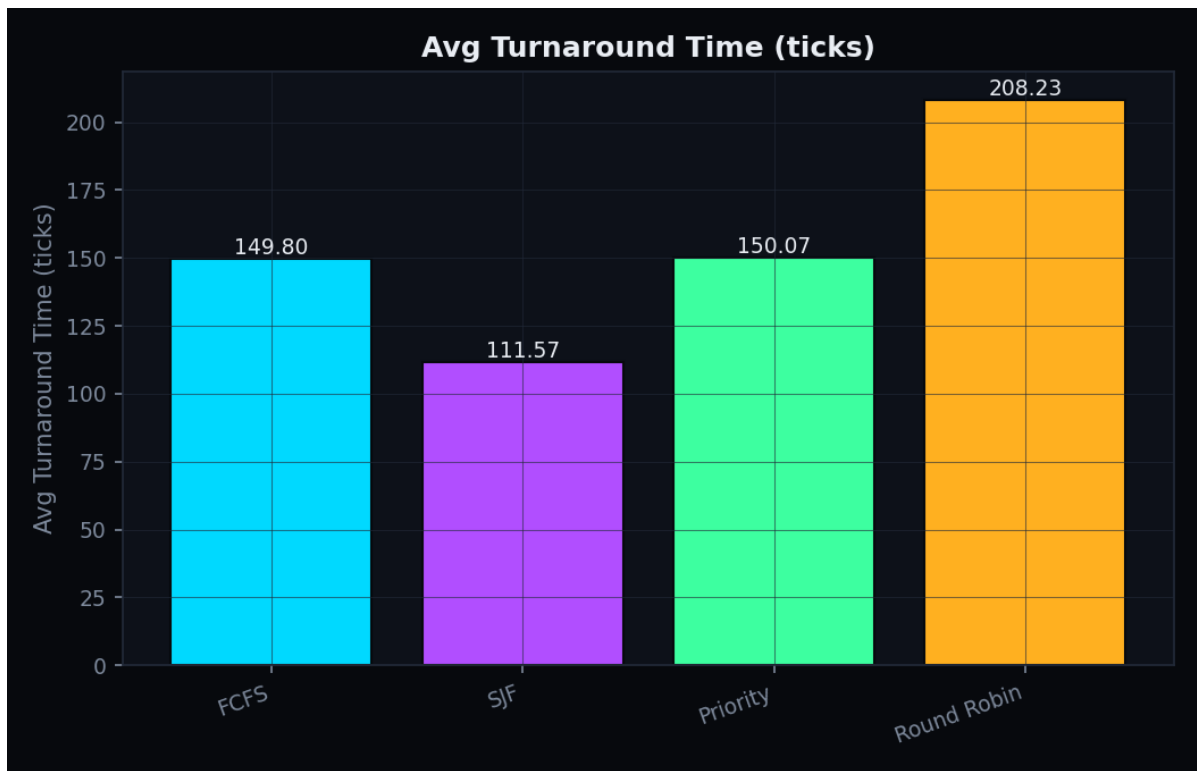
Algorithm	Avg Wait	Avg TAT	Avg Resp	CPU%	Switches	Fairness	Efficiency
FCFS	138.3	149.8	138.3	98.9	29	0.754	37.2
SJF	100.1	111.6	100.1	98.9	29	0.540	17.0
Priority	138.6	150.1	138.6	98.9	29	0.653	29.3
Round Robin	196.7	208.2	54.1	98.9	95	0.860	56.1

Fairness = Jain's Fairness Index (standard). Efficiency = a designed composite heuristic ($100 \times \text{CPU utilization} \times \text{fairness} / (1 + \text{starvation})$), original to this project — not a standard OS metric. See README for full definitions.

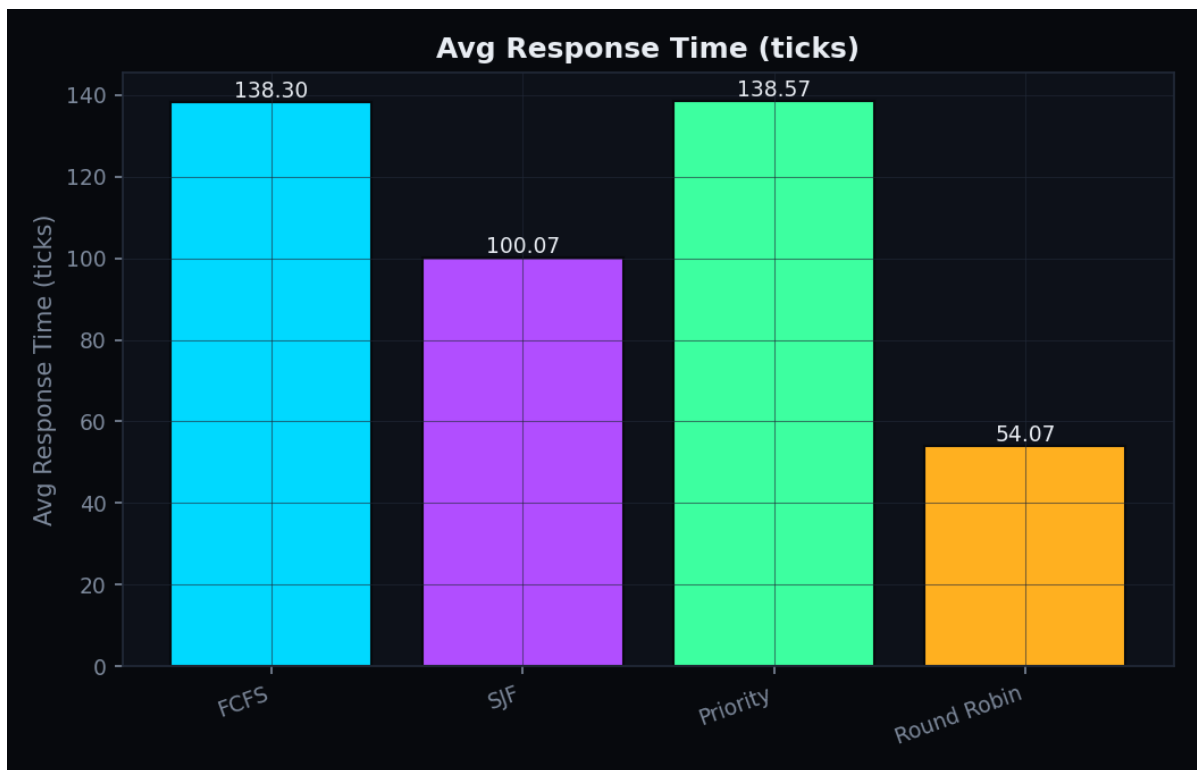
Comparison Charts



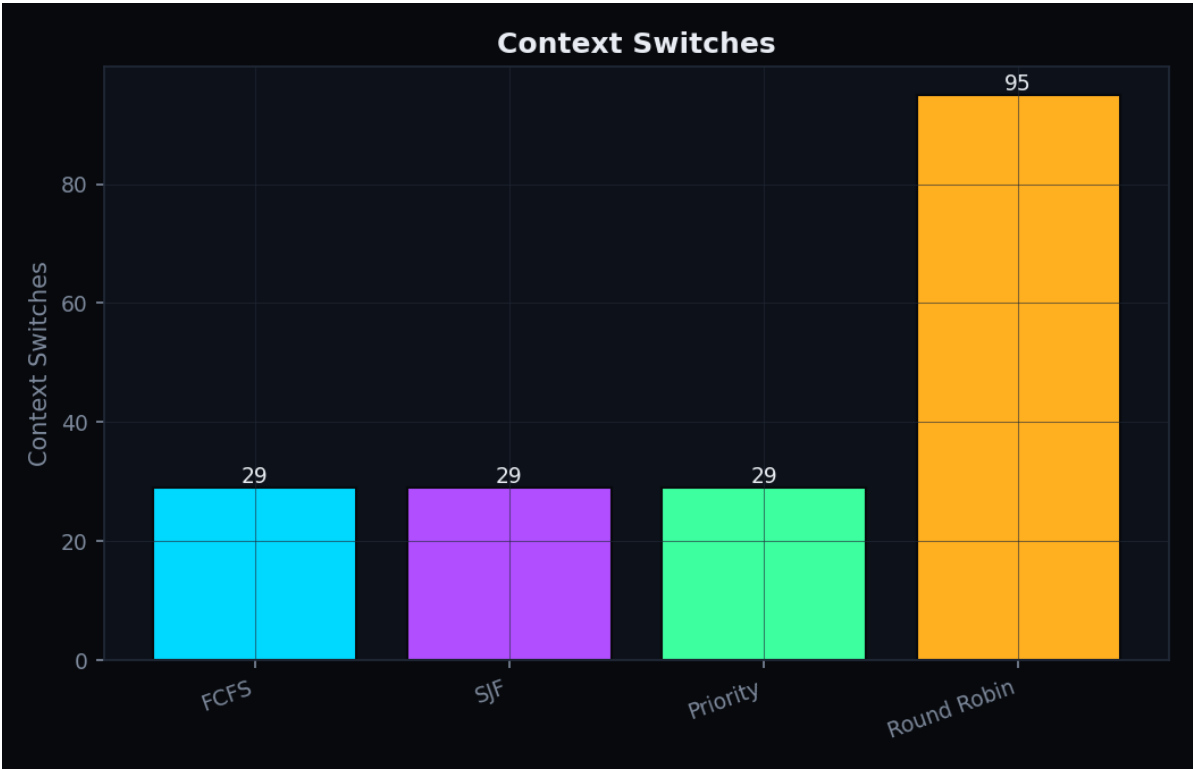
Average Waiting Time by Algorithm



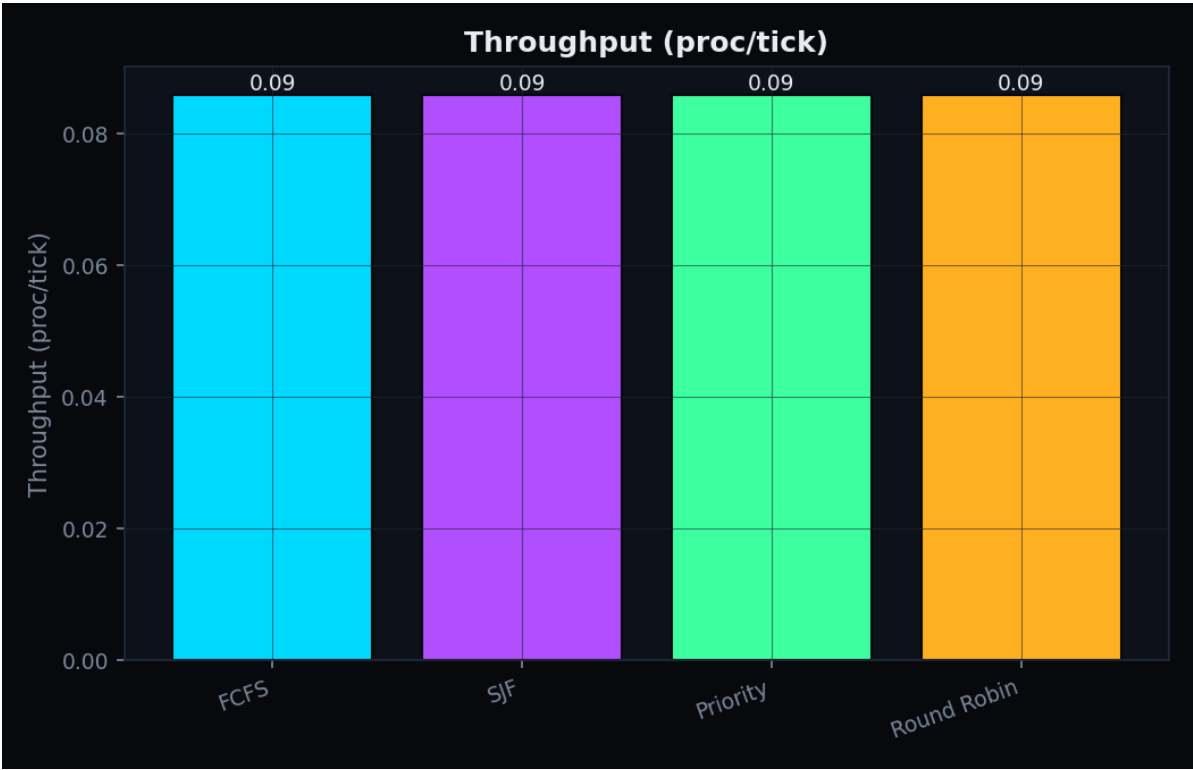
Average Turnaround Time by Algorithm



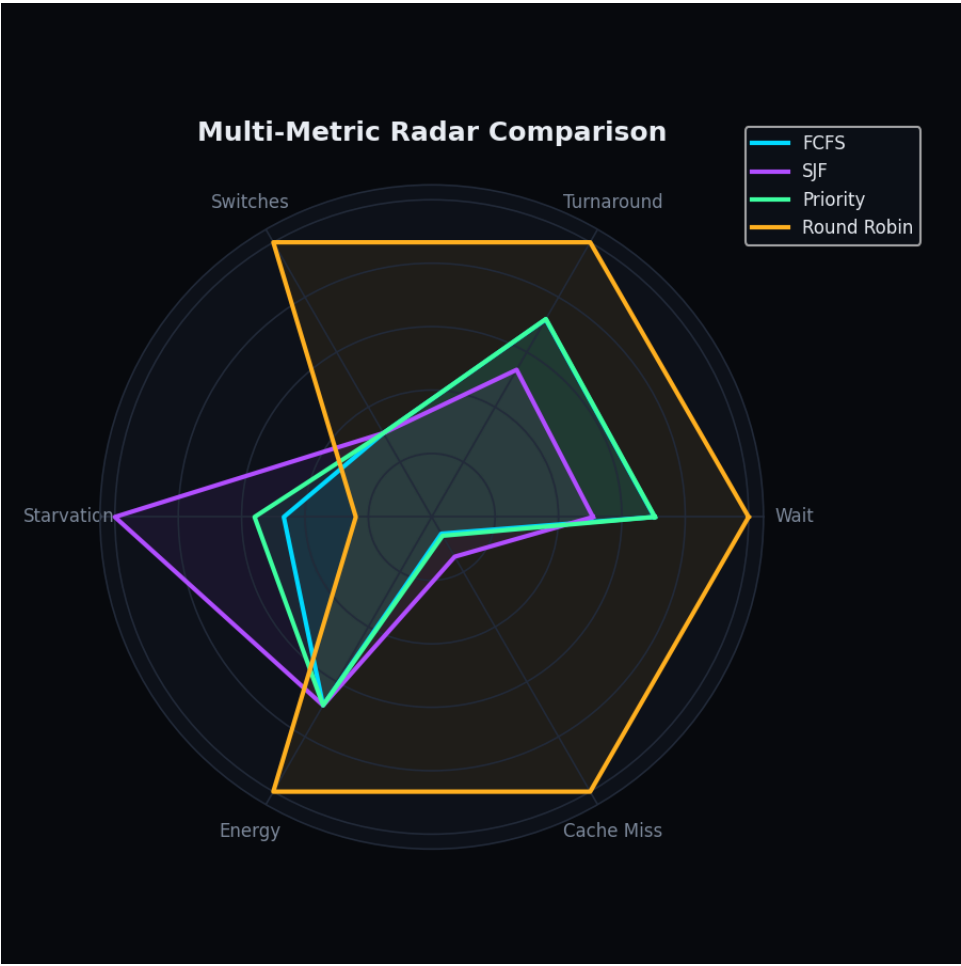
Average Response Time by Algorithm



Context Switches by Algorithm



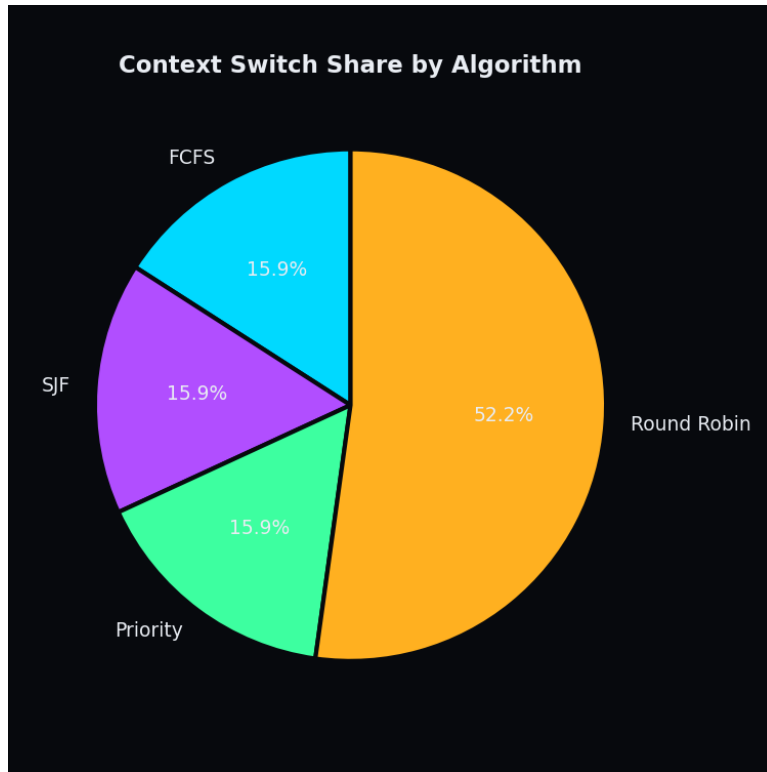
Throughput by Algorithm



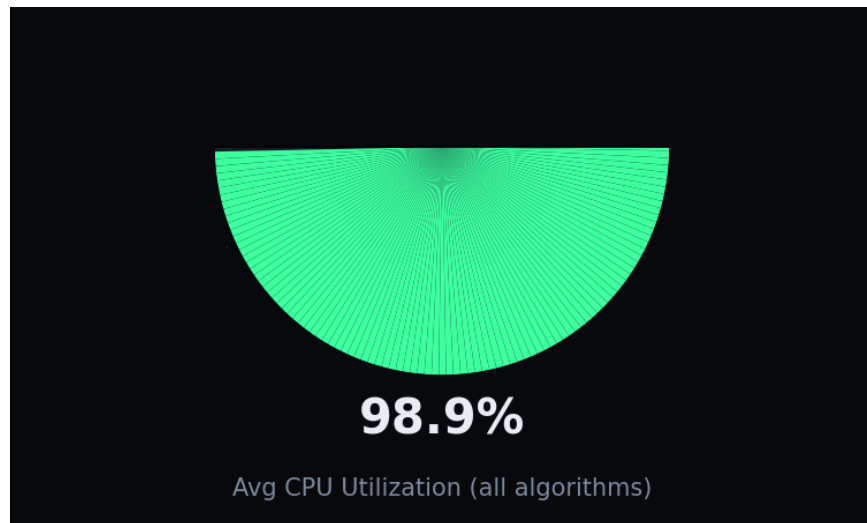
Multi-metric radar comparison (normalized per metric).

Performance Heatmap (green = better)								
FCFS	138.30	149.80	138.30	0.75	1.00	29	0.99	37.20
SJF	100.07	111.57	100.07	0.54	2.15	29	0.99	16.96
Priority	138.57	150.07	138.57	0.65	1.20	29	0.99	29.34
Round Robin	196.73	208.23	54.07	0.86	0.51	95	0.99	56.10
	Avg Wait	Avg TAT	Avg Resp	Fairness	Starvation	Switches	CPU Util	Efficiency

Performance heatmap — green indicates the better value per column.



Share of total context switches contributed by each algorithm.

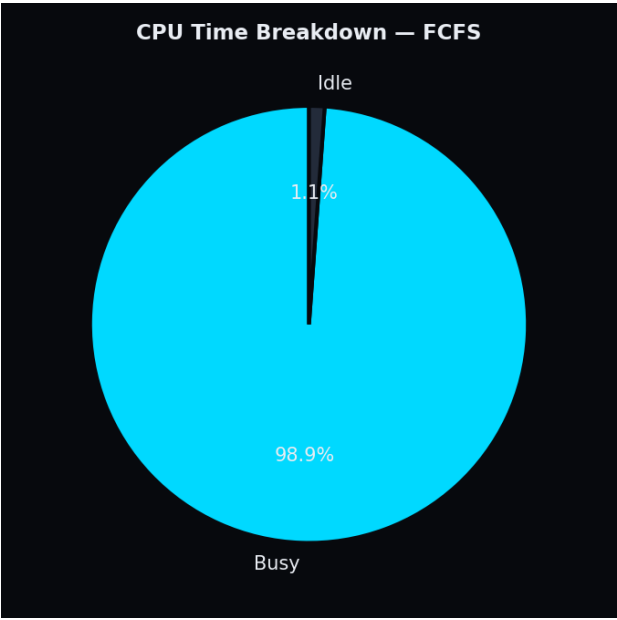


Average CPU utilization across all algorithms in this run.

Detail — FCFS



Execution timeline (Gantt) — FCFS

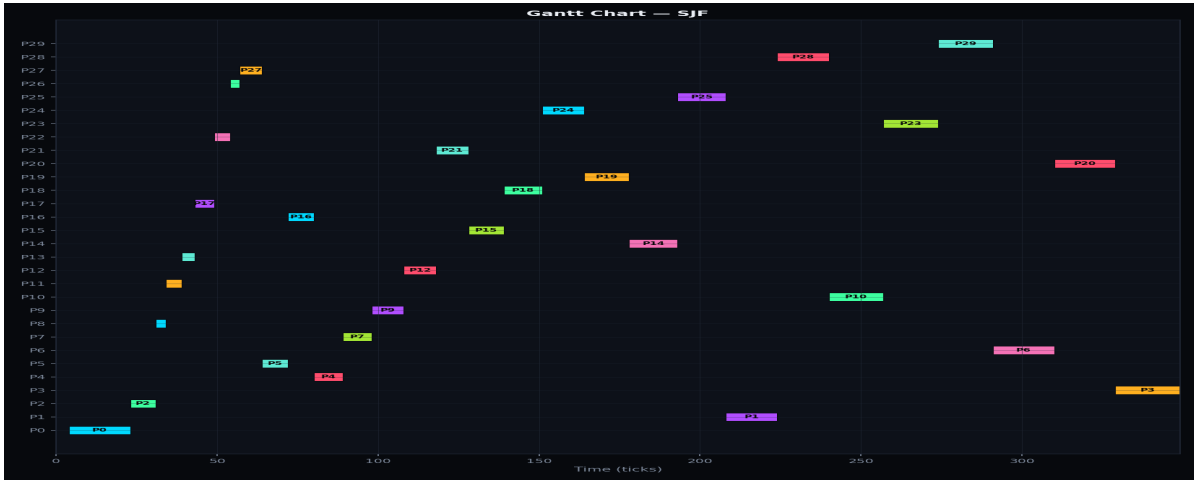


Busy vs. idle CPU time — FCFS

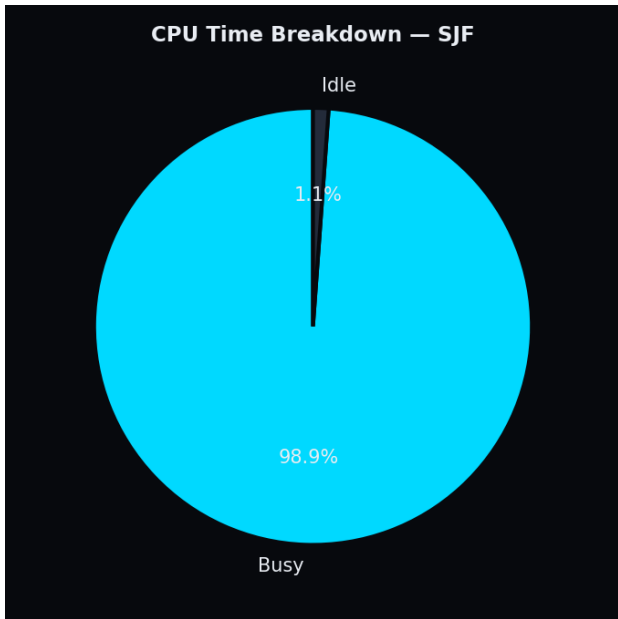
Metric	Value
Avg Waiting Time	138.30 ticks
Avg Turnaround Time	149.80 ticks
Avg Response Time	138.30 ticks
CPU Utilization	98.9%
Idle CPU Time	4 ticks (1.1%)
Throughput	0.0860 proc/tick
Context Switches	29
Fairness Index	0.754
Starvation Score	1.00

Scheduling Efficiency Score	37.2 / 100
Energy Score (simulated)	432 units
Cache Miss Rate (simulated)	5.2%

Detail — SJF



Execution timeline (Gantt) — SJF

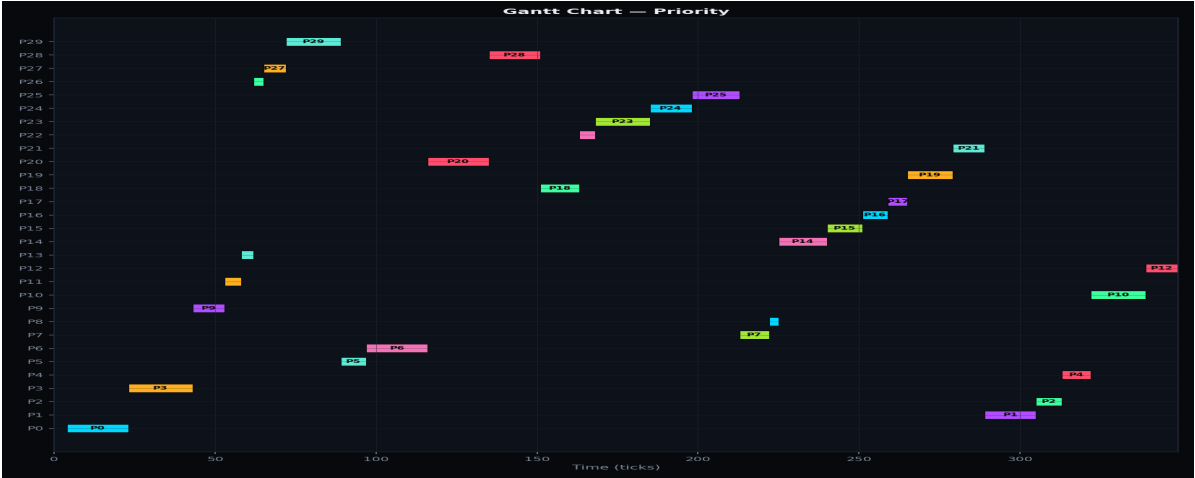


Busy vs. idle CPU time — SJF

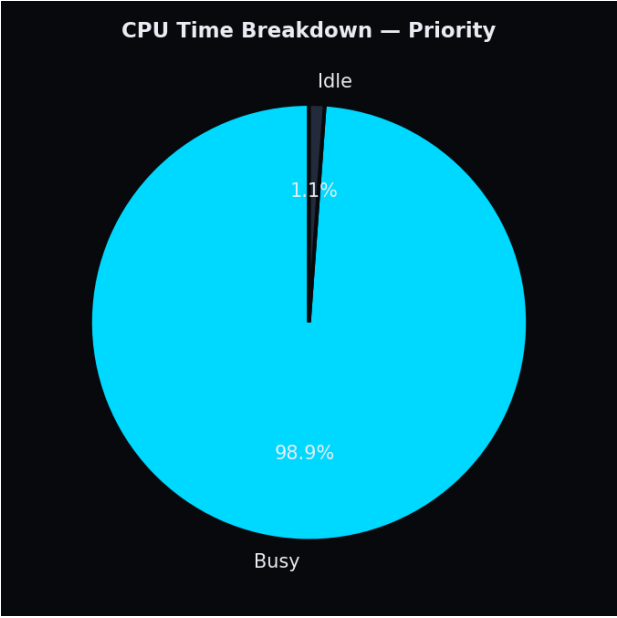
Metric	Value
Avg Waiting Time	100.07 ticks
Avg Turnaround Time	111.57 ticks
Avg Response Time	100.07 ticks
CPU Utilization	98.9%
Idle CPU Time	4 ticks (1.1%)
Throughput	0.0860 proc/tick
Context Switches	29
Fairness Index	0.540
Starvation Score	2.15

Scheduling Efficiency Score	17.0 / 100
Energy Score (simulated)	432 units
Cache Miss Rate (simulated)	12.4%

Detail — Priority



Execution timeline (Gantt) — Priority

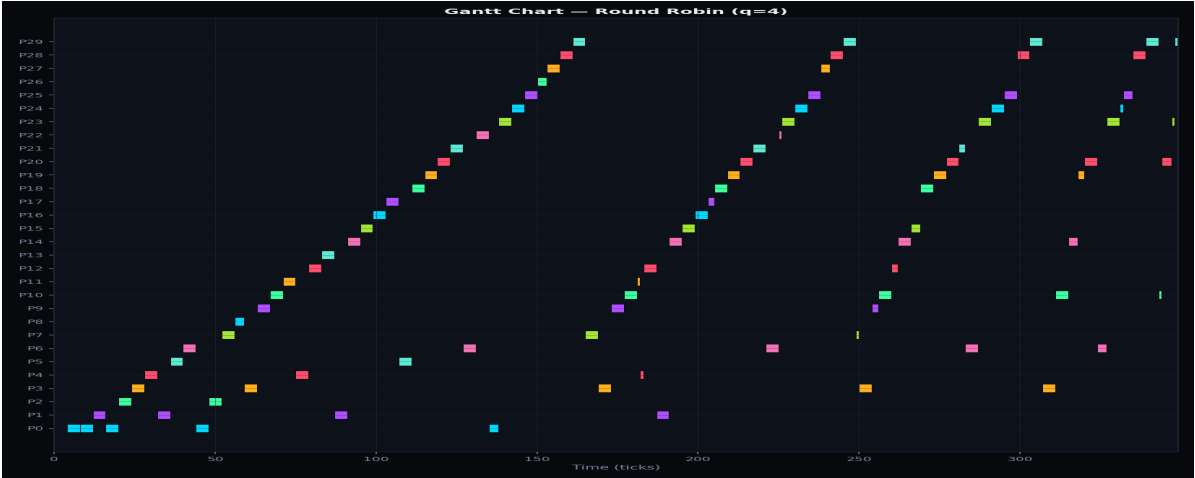


Busy vs. idle CPU time — Priority

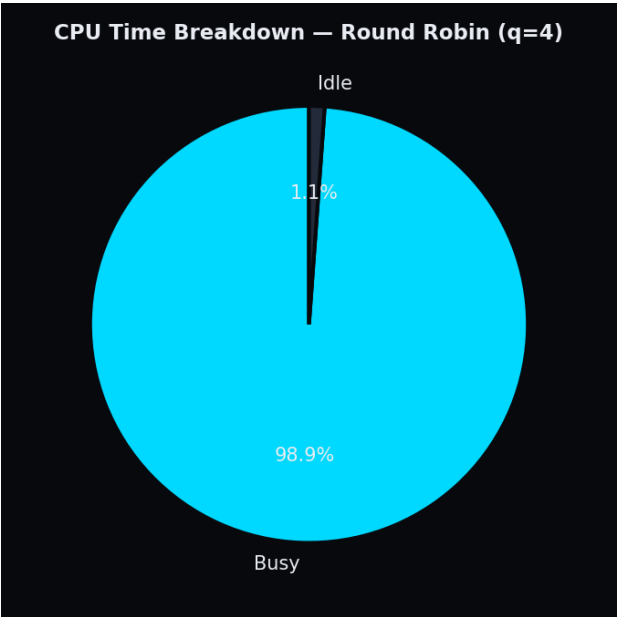
Metric	Value
Avg Waiting Time	138.57 ticks
Avg Turnaround Time	150.07 ticks
Avg Response Time	138.57 ticks
CPU Utilization	98.9%
Idle CPU Time	4 ticks (1.1%)
Throughput	0.0860 proc/tick
Context Switches	29
Fairness Index	0.653
Starvation Score	1.20

Scheduling Efficiency Score	29.3 / 100
Energy Score (simulated)	432 units
Cache Miss Rate (simulated)	5.9%

Detail — Round Robin



Execution timeline (Gantt) — Round Robin



Busy vs. idle CPU time — Round Robin

Metric	Value
Avg Waiting Time	196.73 ticks
Avg Turnaround Time	208.23 ticks
Avg Response Time	54.07 ticks
CPU Utilization	98.9%
Idle CPU Time	4 ticks (1.1%)
Throughput	0.0860 proc/tick
Context Switches	95
Fairness Index	0.860
Starvation Score	0.51

Scheduling Efficiency Score	56.1 / 100
Energy Score (simulated)	630 units
Cache Miss Rate (simulated)	85.6%